Vectors: Components and Operations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

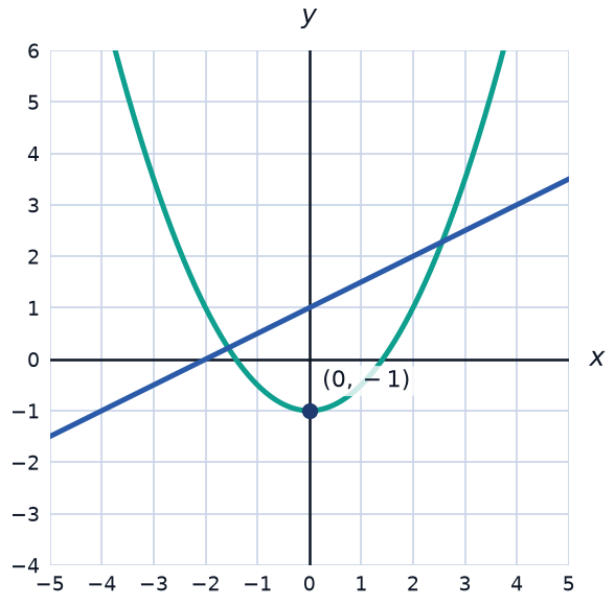
1 [Foundational] • Target: Vector arithmetic

Let $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 4 \rangle$. Find:

a $\mathbf{u} + \mathbf{v}$ **b** $3\mathbf{u} - 2\mathbf{v}$ **c** $|\mathbf{u}|$

2 [Foundational] • Target: Vector arithmetic

Find the component form and magnitude of the vector from $P(1, 3)$ to $Q(5, -2)$.



3 [Foundational] • Target: Applications

A plane flies due east at 200 mph while the wind blows due north at 50 mph.

a Find the magnitude of the plane's resultant velocity, to one decimal place.

b Find the direction of the resultant, as an angle north of east to two decimal places.

4 [Foundational] • Target: Applications

A boat aims to cross a river by heading due north at 4 m/s, but the current pushes it east at 1.5 m/s.

a Find the boat's resultant speed, to two decimal places.

b If the river is 200 m wide, find how long the crossing takes.

c Find how far downstream (east) the boat lands.

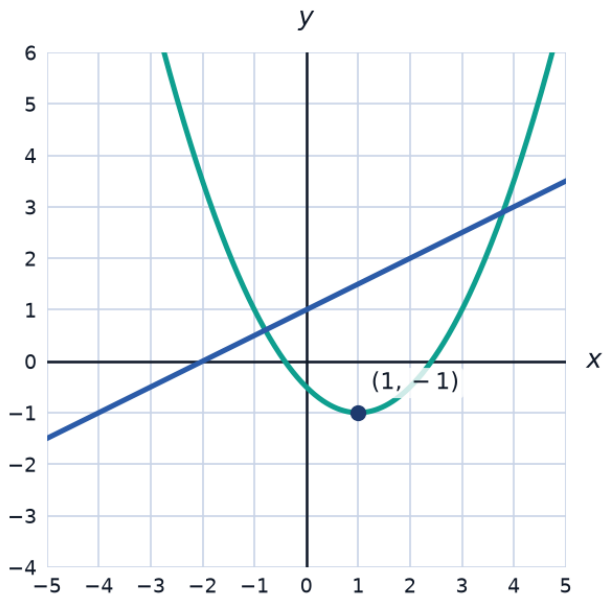
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5 [Foundational] • Target: Position vectors and geometry

Given $A(-2, 5)$ and $B(4, -3)$, find the midpoint of $\overline{AB}$ and the vector $\vec{AB}$.

6 [Foundational] • Target: More applications

Given $\mathbf{u} = \langle 4, -1 \rangle$, find a vector orthogonal to $\mathbf{u}$ with the same magnitude as $\mathbf{u}$.



7 [Foundational] • Target: Finding direction angles

Find the direction angle of $\mathbf{v} = \langle -3, 3\sqrt{3} \rangle$, to the nearest degree.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Parallel and equal vectors

Determine whether $\mathbf{u} = \langle 4, -6 \rangle$ and $\mathbf{v} = \langle -6, 9 \rangle$ are parallel, and if so, state whether they point in the same or opposite directions.

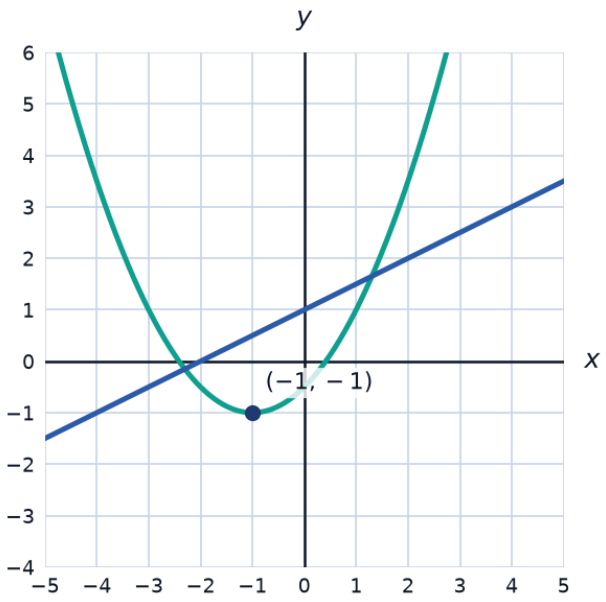
9 [Intermediate] • Target: Vectors in three real-world scenarios

A hiker walks 3 km east, then 4 km north, then 2 km west. Find the magnitude of the hiker's total displacement from the starting point.

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10 [Intermediate] • Target: Vectors in three real-world scenarios

Three ropes pull on a ring with forces $\langle 10, 0\rangle$, $\langle -5, 8.66\rangle$, and $\langle -5, -8.66\rangle$ N. Find the net force, and explain what this implies physically.



11 [Intermediate] • Target: Vectors in three real-world scenarios

A crane hoists a load using two cables. One cable exerts a force of $\langle -200, 350\rangle$ N and the other exerts $\langle 200, 400\rangle$ N. Find the combined lifting force (the vertical component of the resultant), and explain why the horizontal components matter for the crane's structural design even though they cancel.

12 [Intermediate] • Target: Vectors in three real-world scenarios

A swimmer aims to swim directly across a river at 1.5 m/s relative to the water, while the current flows at 0.8 m/s downstream. Find the swimmer's resultant speed relative to the shore, and the angle (from the intended straight-across direction) at which they actually drift, to the nearest degree.

13 [Intermediate] • Target: Vector arithmetic

Find the unit vector in the direction of $\mathbf{v} = \langle 6, -8\rangle$.

14 [Intermediate] • Target: Vector arithmetic

Write the vector of magnitude 12 making an angle of $150°$ with the positive x-axis in component form (exact values).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

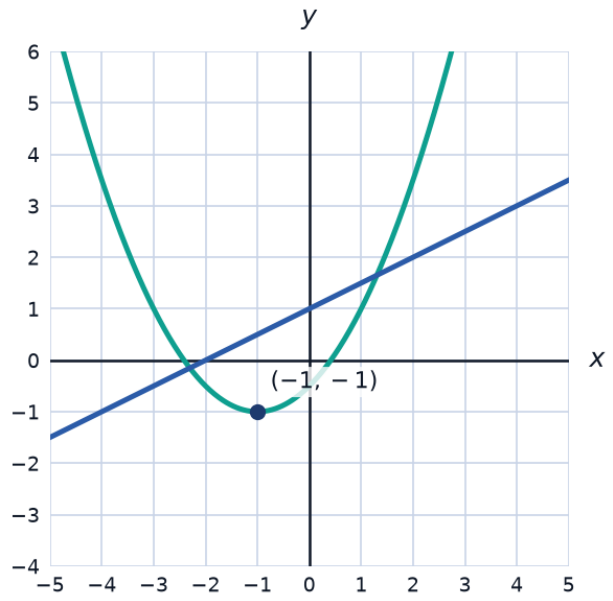
15 [Challenging] • Target: Applications • Cross-Lesson: 1.01 Functions and Function Notation

Two forces $\mathbf{F}_1 = \langle 30, 0\rangle$ N and $\mathbf{F}_2 = \langle 0, 40\rangle$ N act on an object. Find the magnitude of the resultant force.

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16 [Challenging] • Target: Position vectors and geometry • Cross-Lesson: 1.02 Domain and Range

Let $\mathbf{a} = \langle 2, 3 \rangle$ and $\mathbf{b} = \langle -4, 1 \rangle$. Find a vector $\mathbf{c}$ such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \langle 0, 0 \rangle$, and verify your answer.



17 [Challenging] • Target: Position vectors and geometry • Cross-Lesson: 1.03 Average Rate of Change

Express $\mathbf{v} = \langle 5, -12 \rangle$ in terms of the standard unit vectors $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{v}|$.

18 [Challenging] • Target: More applications • Cross-Lesson: 2.01 Right-Triangle Trigonometry

An airplane has an airspeed of 305 mph and flies on a heading $7.5°$ west of due north, while a 40 mph wind blows due east. Find the plane's resultant ground velocity vector, and its ground speed, to one decimal place (take east as the positive x-direction and north as the positive y-direction).

19 [Challenging] • Target: Finding direction angles • Cross-Lesson: 2.02 Radian Measure and Arc Length

A vector has magnitude 10 and direction angle $200°$. Write it in component form, rounded to two decimal places.

20 [Challenging] • Target: Vector equations and unknowns • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Find scalars c_1 and c_2 such that $c_1\langle 1, 0 \rangle + c_2\langle 0, 1 \rangle = \langle -3, 7 \rangle$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Geometric vs Component Form

Find the component form of vector $\mathbf{v}$ with initial point $P(2, -3)$ and terminal point $Q(-1, 4)$.

a. $\langle -3, 7 \rangle$

b. $\langle 1, 1 \rangle$

c. $\langle 3, -7 \rangle$

d. $\langle -2, -12 \rangle$

22 [Foundational] • Target: Geometric vs Component Form

Express $\mathbf{v} = \langle 5, -8 \rangle$ in terms of the standard unit basis vectors $\mathbf{i}$ and $\mathbf{j}$.

a. $5\mathbf{i} - 8\mathbf{j}$

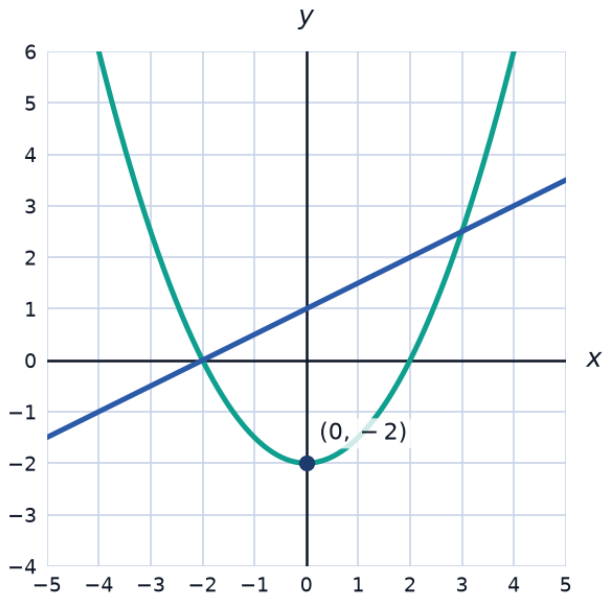
b. $5\mathbf{i} + 8\mathbf{j}$

c. $-8\mathbf{i} + 5\mathbf{j}$

d. $5\mathbf{j} - 8\mathbf{i}$

23 [Foundational] • Target: Geometric vs Component Form

When are two geometric vectors considered equal?



a. When they have both the same magnitude and the same direction

b. When they have the same initial point

c. When their lengths are equal regardless of direction

d. When they are collinear

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24 [Foundational] • Target: Magnitude and Direction Angle

Find the magnitude of $\mathbf{v} = \langle -6, 8 \rangle$.

a. 10

b. 14

c. $\sqrt{28}$

d. 2

25 [Foundational] • Target: Magnitude and Direction Angle

Find the direction angle $\theta \in [0^\circ, 360^\circ)$ of $\mathbf{v} = \langle -3, 3\sqrt{3} \rangle$.

a. 120°

b. 60°

c. 150°

d. 240°

26 [Foundational] • Target: Magnitude and Direction Angle

Find the components of a vector with magnitude 12 and direction angle 210°.

a. $\langle -6\sqrt{3}, -6 \rangle$

b. $\langle -6, -6\sqrt{3} \rangle$

c. $\langle 6\sqrt{3}, 6 \rangle$

d. $\langle -6\sqrt{2}, -6\sqrt{2} \rangle$

27 [Foundational] • Target: Vector Addition, Subtraction, and Scalar Multiplication

Given $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 5 \rangle$, find $2\mathbf{u} - 3\mathbf{v}$.

a. $\langle 9, -19 \rangle$

b. $\langle 3, 11 \rangle$

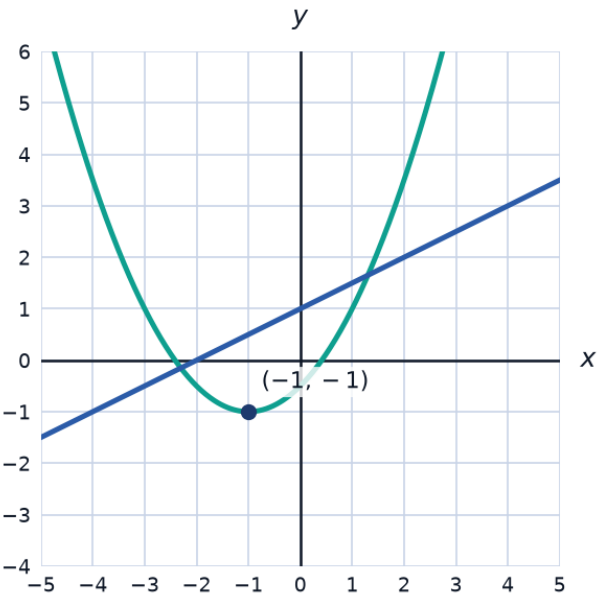
c. $\langle 9, 11 \rangle$

d. $\langle 3, -19 \rangle$

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28 [Foundational] • Target: Vector Addition, Subtraction, and Scalar Multiplication

Find the unit vector $\mathbf{u}$ in the same direction as $\mathbf{v} = \langle 5, -12 \rangle$.



a. $\langle 5/13, -12/13 \rangle$

b. $\langle 5/17, -12/17 \rangle$

c. $\langle 1, -1 \rangle$

d. $\langle 5, -12 \rangle$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Vector Addition, Subtraction, and Scalar Multiplication

What is the result of multiplying any vector $\mathbf{v}$ by scalar -1?

a. A vector of the same magnitude pointing in the exactly opposite direction

b. The zero vector

c. A scalar number

d. A perpendicular vector

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30 [Intermediate] • Target: Resultant Forces and Navigation

Two forces act on an object: $\mathbf{F}_1 = \langle 20, 0 \rangle$ N and $\mathbf{F}_2 = \langle 0, 15 \rangle$ N. Find the magnitude of the resultant force.

a. 25 N

b. 35 N

c. 5 N

d. 300 N

31 [Intermediate] • Target: Resultant Forces and Navigation

An airplane flies at a heading due east with airspeed 400 mph. A wind blows from the south at 50 mph. Find the ground speed of the plane.

a. $\sqrt{400^2 + 50^2} \approx 403.1$ mph

b. 450 mph

c. 350 mph

d. 425 mph

32 [Intermediate] • Target: Resultant Forces and Navigation

In the airplane problem, what is the drift angle north of east?

a. $\arctan(50/400) \approx 7.1^\circ$

b. 12.5°

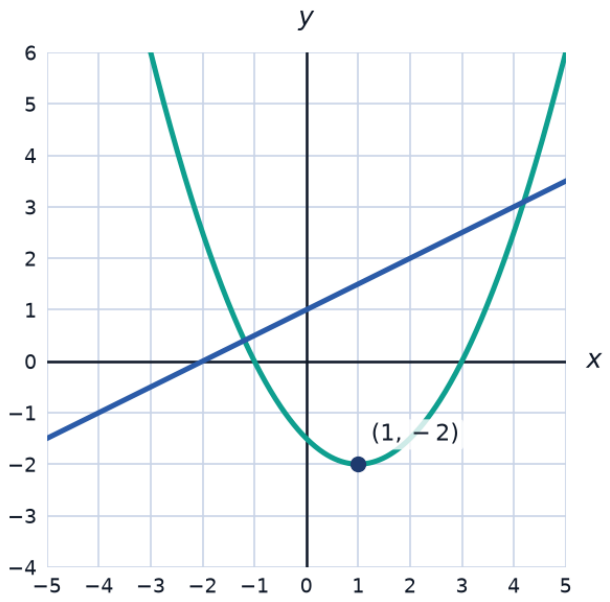
c. 45°

d. 3.6°

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33 **[Intermediate]** • **Target: Vector arithmetic**

Consider two points in the Cartesian plane, $P = (2, 3)$ and $Q = (5, 7)$. What is the component form of the vector $\vec{PQ}$?



a. $\langle 3, 4 \rangle$

b. $\langle -3, -4 \rangle$

c. $\langle 7, 10 \rangle$

d. $\langle 1, 4 \rangle$

34 **[Intermediate]** • **Target: Applications**

Find the unit vector in the direction of $\mathbf{v} = \langle -8, 15 \rangle$.

a. $\left\langle \frac{9}{17}, \frac{15}{17} \right\rangle$

b. $\left\langle \frac{15}{17}, -\frac{8}{17} \right\rangle$

c. $\left\langle -\frac{8}{17}, \frac{15}{17} \right\rangle$

d. $\langle -8, 15 \rangle$

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35 [Intermediate] • Target: Position vectors and geometry

Find the unit vector in the direction of $\mathbf{v} = \langle -6, 8 \rangle$.

a. $\left\langle \frac{2}{5}, \frac{4}{5} \right\rangle$

b. $\langle -6, 8 \rangle$

c. $\left\langle -\frac{3}{5}, \frac{4}{5} \right\rangle$

d. $\left\langle \frac{4}{5}, -\frac{3}{5} \right\rangle$

36 [Intermediate] • Target: More applications

Find the unit vector in the direction of $\mathbf{v} = \langle 3, -4 \rangle$.

a. $\left\langle \frac{8}{5}, -\frac{4}{5} \right\rangle$

b. $\langle 3, -4 \rangle$

c. $\left\langle -\frac{4}{5}, \frac{3}{5} \right\rangle$

d. $\left\langle \frac{3}{5}, -\frac{4}{5} \right\rangle$

37 [Intermediate] • Target: Finding direction angles

Consider two vectors, $\mathbf{u} = (3; k)$ and $\mathbf{v} = (-2; 4)$. If $\mathbf{u}$ is parallel to $\mathbf{v}$, find the value of the scalar parameter k.

a. -6

b. 6

c. 3

d. -3

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Vector equations and unknowns • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Consider the vector $\mathbf{v} = \langle 3, 4 \rangle$. What is the magnitude of this vector?

a. 5

b. 7

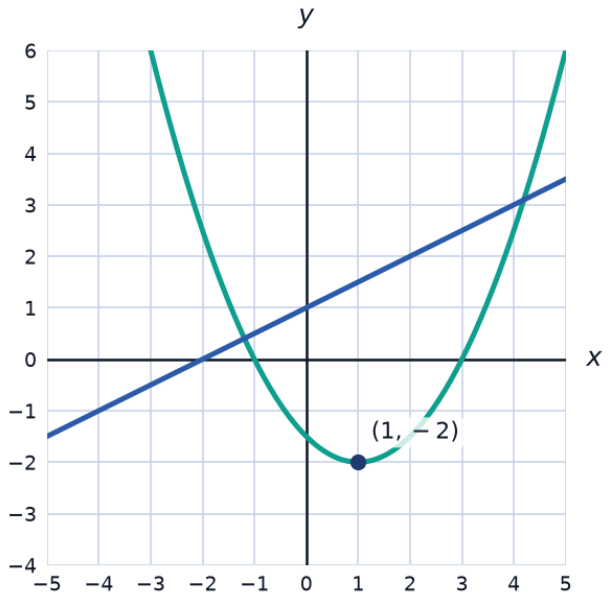
c. 12

d. 25

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39 [Challenging] • Target: Parallel and equal vectors • Cross-Lesson: 2.02 Radian Measure and Arc Length

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. What are the components of the vector $\mathbf{v}$?



a. ■3, 4■

b. ■7, 10■

c. ■-3, -4■

d. ■4, 3■

40 [Challenging] • Target: Vectors in three real-world scenarios • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Find the angle between $\mathbf{u} = \langle 1, 3 \rangle$ and $\mathbf{v} = \langle 2, -2 \rangle$ (to one decimal place).

a. 116.6 ˚

b. 106.6 ˚

c. 126.6 ˚

d. 63.4 ˚

41 [Challenging] • Target: Geometric vs Component Form • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Find the angle between $\mathbf{u} = \langle 4, 1 \rangle$ and $\mathbf{v} = \langle 3, -1 \rangle$ (to one decimal place).

a. 147.5 ˚

b. 42.5 ˚

c. 22.5 ˚

d. 32.5 ˚

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42 [Challenging] • Target: Magnitude and Direction Angle • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Find the angle between $\mathbf{u} = \langle 3, 2 \rangle$ and $\mathbf{v} = \langle 2, 2 \rangle$ (to one decimal place).

a. 1.3°

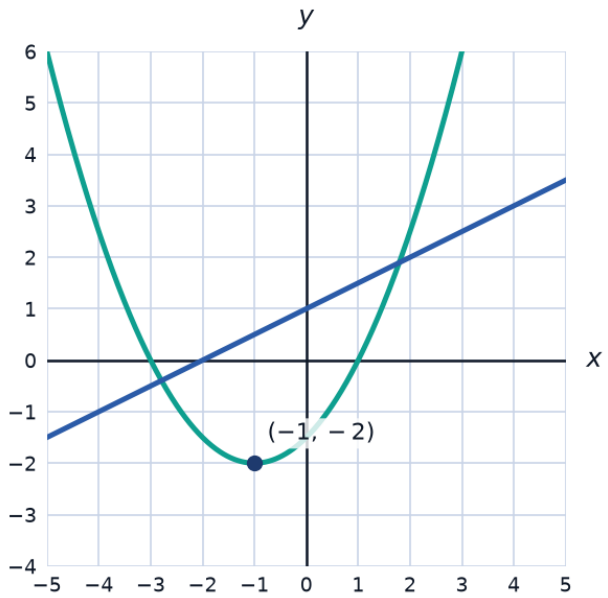
b. 21.3°

c. 11.3°

d. 168.7°

43 [Challenging] • Target: Vector Addition, Subtraction, and Scalar Multiplication • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the angle between $\mathbf{u} = \langle 2, 3 \rangle$ and $\mathbf{v} = \langle 2, 2 \rangle$ (to one decimal place).



a. 1.3°

b. 21.3°

c. 168.7°

d. 11.3°

44 [Challenging] • Target: Resultant Forces and Navigation • Cross-Lesson: 3.03 Sum and Difference Formulas

Find the angle between $\mathbf{u} = \langle 3, 3 \rangle$ and $\mathbf{v} = \langle 3, 2 \rangle$ (to one decimal place).

a. 168.7°

b. 21.3°

c. 1.3°

d. 11.3°

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45 [Challenging] • Target: Vector arithmetic • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Find the angle between $\mathbf{u} = \langle 3, 3 \rangle$ and $\mathbf{v} = \langle 4, 1 \rangle$ (to one decimal place).

a. 31.0°

b. 41.0°

c. 21.0°

d. 149.0°

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